

Ibrahim Al Azher

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AREA OF EXPERTISE

AI Researcher specializing in Agentic AI and Large Language Model post-training, with a focus on collaborative multi-agent systems and tool-augmented reasoning. Expertise in optimizing LLMs using reinforcement learning techniques (SFT, RLHF, DPO, GRPO) with verifiable reward functions in distributed HPC environments. Demonstrated success in building secure, robust AI architectures that defend against jailbreaking and backdoor attacks. Experienced in applying sophisticated RAG and multimodal pipelines to solve complex problems across diverse domains, including scientific document processing, software vulnerability detection, healthcare, and federated learning for financial fraud detection.

EDUCATION

Northern Illinois University

PhD Computer Science; GPA: 3.89

Dekalb, IL

August 2023 - December 2027 (tentative)

Northern Illinois University

M.S Computer Science; GPA: 3.72

Dekalb, IL

August 2023 - May 2025

Shahjalal University of Science and Technology

B.Sc. Computer Science; GPA: 3.69

Sylhet, Bangladesh

March 2016 - March 2021

EXPERIENCE

Research Assistant

Northern Illinois University

Aug 2023 - Present

Dekalb, IL

- Designed and evaluated multi-agent LLM systems for reliable reasoning and tool use, using shared memory, MCP-based external tools, and iterative feedback loops.
- Built RAG pipelines with hybrid retrieval, semantic search, reranking, and knowledge graph integration for evidence-grounded generation.
- Developed evaluation frameworks to measure accuracy, relevance, grounding, and latency for LLM-generated outputs and retrieval pipelines.
- Worked on LLM post-training and optimization, including SFT, DPO, GRPO, and RLHF-style workflows, for improving multi-agent generation quality.
- Contributed to writing of an NSF, DARPA proposal.

Software Engineer

BYSL Global Technology Group

June 2021 - Feb 2022

Dhaka, Bangladesh

- Works on scalable projects from design to coding, testing, and installation using Python, XML, Javascript, PostgreSQL, and jQuery in Linux OS.
- Linux OS. Designed and developed various ERP-based business software and modules like sales, inventory, purchase, website, and e-commerce using Python language in Odoo.

Junior Software Engineer

Unisoft Systems Limited

December 2020 - May 2021

Dhaka, Bangladesh

- Research, assess, and lead the initiation of new technologies to maximize performance. Interact with clients and analyze requirements and develop an application in Python.
- Had written various SQL to generate numerous reports and created various modules using the OOP, Database model.

Published

- **Ibrahim Al Azher**, Miftahul Jannat Mokarrama, Zhishuai Guo, Sagnik Ray Choudhury, Hamed Alhoori. “BAGELS: Benchmarking the Automated Generation and Extraction of Limitations from Scholarly Text”. **Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025, China**. Introduced an LLM-RAG-based framework, effectively extracts, generates from papers and relevant papers, proposes a new evaluation framework for Limitation Generation from scientific articles. [Paper](#). [Code](#). [Dataset](#).
- **Ibrahim Al Azher**, Venkata Devesh Reddy Seethi, Akhil Pandey Akella, and Hamed Alhoori. “LimTopic: LLM-based Topic Modeling and Text Summarization for Analyzing Scientific Articles’ Limitations.” . 2024 ACM/IEEE Joint Conference on Digital Libraries, December 2024, Hong Kong. Introduced LimTopic, a strategy for using topic generation with meaningful titles and descriptions in limitation sections in scientific articles with Topic Modeling and Large Language Models (LLMs). [Paper](#). [Arxiv](#). [Code](#).
- **Ibrahim Al Azher**, Miftahul Jannat Mokarrama, Zhishuai Guo, Sagnik Ray Choudhury, Hamed Alhoori “FutureGen: A RAG-based Approach to Generate the Future Work of Scientific Article”. International Workshop on AI for Scientific Communication (Ai4SC 2025), IEEE eScience 2025, Chicago. An LLM-based RAG approach to generate future work for scientific articles with a self-feedback approach. [Paper](#)
- **Ibrahim Al Azher**, Zhishuai Guo, and Hamed Alhoori. “LimAgents: Limitation Generation of Scientific Article with LLM Agents and RAG”. ACM/IEEE Joint Conference on Digital Libraries, 2026, Texas. Generating scientific article limitations using multiple LLM Agents, such as an extractor, analyzer, reviewer, and citation. A judge will evaluate and provide feedback to improve performance. [Paper](#). [Code](#). [Dataset](#)
- **Ibrahim Al Azher**, and Hamed Alhoori. “Towards Automated Analysis of Scientific Limitations and FutureWork.” ACM/IEEE Joint Conference on Digital Libraries 2026, Dallas, Texas.
- **Ibrahim Al Azher**, and Hamed Alhoori. “FutureLens: Longitudinal Topic Patterns in Scholarly Text and Future-Work References ” AI4SciSci workshop, ACM/IEEE Joint Conference on Digital Libraries 2026, Dallas, Texas.
- **Ibrahim Al Azher**, and Hamed Alhoori. “Generating Suggestive Limitations from Research Articles Using LLM and Graph-Based Approach.” 2024 ACM/IEEE Joint Conference on Digital Libraries, December 2024, Hong Kong. [Paper](#)
- **Ibrahim Al Azher**, and Hamed Alhoori. “Mitigating Visual Limitations of Research Papers.” 2024 IEEE International Conference on Big Data (IEEE BigData 2024), December 2024, Washington D.C. Visual limitations in scientific research papers, such as charts, graphs, and diagrams, are addressed with a multimodal LLM (QWen, Llava, Llama, and GPT-4) to generate clear, concise descriptions incorporating LLM feedback and LLM as a judge for evaluation. [Paper](#). [PDF](#).
- **Ibrahim Al Azher**, Hamed Alhoori. “Generating Suggestive Limitations from Research Articles Using LLM and Graph-Based Approach” The 11th IEEE International Conference on Data Science and Advanced Analytics (DSAA 2024), October 2024, San Diego, CA. A graph-based approach to recommend limitations of a paper by creating a knowledge graph using citations and the semantic relations of papers, incorporating the subgraph into LLM.
- **Ibrahim Al Azher**, and Hamed Alhoori. “Trend Prediction of Scientific Article Future Work.” A topic modeling-based trend prediction of scientific articles by analyzing future work. 2025 IEEE International Conference on Big Data. Macau SAR, China
- **Ibrahim Al Azher**, Sohel Ahmed, and Md. Saiful Islam. “Identifying Author in Bengali Literature by Bi-LSTM with Attention Mechanism.” 24th International Conference on Computer and Information Technology (ICCIT), December 2021. [Paper](#). [PDF](#)
- Tamzid Azad, **Ibrahim Al Azher**, Sagnik Ray Choudhury, and Hamed Alhoori. “Predicting Scholarly Impact with Retrieval-Augmented LLMs” Association of Computational Linguistic (ACL), Scholarly Document Processing (SDProc), 2025, Vienna, Austria. [Paper](#).
- Murtuza Shahzad, Joey Wilson, **Ibrahim Al Azher**, Hamed Alhoori, and Mona Rahimi. “From Theory to Practice: Code Generation Using LLMs for CAPEC and CWE Frameworks” 2nd International Workshop on Large Language Models for Code (LLM4Code 2025), ICSE 2025. [Paper](#).
- Christy Muasher-Kerwin, M Courtney Hughes, Michelle Foster, **Ibrahim Al Azher**, Hamed Alhoori. “Exploring Large Language Models for Summarizing and Interpreting an Online Brain Tumor Support Forum.” Sage Journal. [Paper](#)

- MD Saifur Rahman Chowdhury, Nahiyan Habib Khan, Dip Singha, **Ibrahim Al Azher**, Tufael Ahmed, Geerbani Pal Shashi. “Leveraging Self-Sovereign Identity (SSI) with Hyperledger Indy: A Decentralized Identity Ecosystem for Secure Document Management in Bangladesh.” 27th International Conference on Computer and Information Technology (ICCIT) 20-22 December 2024, Cox’s Bazar, Bangladesh. [Paper](#)
- Harish Verma, Miftahul Jannat Mokarrama, **Ibrahim Al Azher** and Hamed Alhoori. “A Comparative Study of ORKG and LLM identified Research Contributions” 2nd Workshop on Innovation Measurement for Scientific Communication (IMSC) in the Era of Big Data, JCDL, December 2024, Hong Kong.
- **Preprint**
 - **Ibrahim Al Azhar**, and Hamed Alhoori. “A Survey on Retrieval Augmented Generation Evaluation Framework”. A systematic review of the methodologies for evaluating RAG systems, which covers evaluation of the retriever, the generator (using NLP-based metrics and the ‘LLM-as-a-Judge’ approach), and the end-to-end system, alongside benchmark datasets and current limitations and future research directions. [PDF](#)
 - **Ibrahim Al Azher**. “Limitations of Scientific Articles and Navigated Future Directions with LLM and RAG.” Masters Thesis. Northern Illinois University, May 2025. [Paper](#)
- **In Progress and Submitted**
 - **Ibrahim Al Azhar**, Sagnik Ray Choudhury, and Hamed Alhoori. “Attacking and Robust defense system in scientific article peer review system”. We investigated jailbreaking, prompt injection, and backdoor attacks in the peer review system and built a robust defense mechanism with a granular evaluation framework.
 - **Ibrahim Al Azhar**, Zhishuai Guo, Sagnik Ray Choudhury and Hamed Alhoori. PolicyGuard Multi Agent Framework for Medical AI Safety and Security Clinical AI models face critical risks from hidden research biases and adversarial attacks. To mitigate this, we engineered an auditable agent LLM pipeline that extracts scientific flaws, models threat vectors, and deploys robust system defenses. The architecture utilizes dual tier RAG for evidence retrieval and aligns local models using step wise Process Rewards and DPO to ensure strict factual accuracy. Extracted vulnerabilities and simulated attacks are processed through a Monte Carlo engine integrated with CDC and Census data to calculate the exact societal harm and financial costs of compromised AI deployments. Awarded 2nd Place at Ignite64 Hackathon and USAII Global AI Finalist. [Code](#) | [YouTube](#)
 - **Ibrahim Al Azhar**, Zhishuai Guo, and Hamed Alhoori. A multi-agent framework for automated limitation generation.” This project introduces a collaborative multi-agent system designed to generate research limitations. The agents leverage access external tools, store the trajectories by rollout, fine tune the agents with reasoning with DPO and GRPO via a custom reward function. To ensure grounding, the framework utilizes a large-scale RAG database, incorporating OpenAlex, cited papers, and a knowledge graph, supported by a hybrid retrieval system and advanced re-ranking mechanisms. [Code](#) | [Live Website](#)
 - **Ibrahim Al Azhar**, Zhishuai Guo, Sagnik Ray Choudhury and Hamed Alhoori. Scientific literature synthesis training with RL with verifiable custom reward function in open ended question answer generation among science.
 - Tamzid Azad, **Ibrahim Al Azher**, Sagnik Ray Choudhury, and Hamed Alhoori. “Scientific Knowledge Transfer with LLMs: Recommending Methods Across Disciplines.”
 - Tamzid Azad, **Ibrahim Al Azher**, and Hamed Alhoori. “How Large Language Models (LLM) affect the Readability Score in Scientific article. ”
 - Bhanu Kedhar **Ibrahim Al Azher**, and Hamed Alhoori. “Benchmarking and classification of Scientific articles figures”

SKILLS

Programming Languages: C, C++, Python, Java, PHP, SQL, Bash/Shell scripting

Large Language Models (LLMs): Zero-shot, few-shot, chain-of-thought, graph-of-thought, tree-of-thought, self-consistency, majority voting, diffusion-based reasoning, Transformers, Mamba; models including GPT, Gemini, Llama, Mistral, and Qwen; logical reasoning, mathematical reasoning, structured reasoning, and multimodal reasoning.

LLM Training & Optimization: Pre-training, instruction tuning, alignment, imitation learning, reinforcement learning, post-training (SFT, RLHF, RLAIF, DPO, GRPO), custom reward design, policy

optimization, parameter-efficient fine-tuning (PEFT, LoRA, QLoRA), quantization, policy distillation, sparse attention, sparsity, KV cache, FlashAttention, inference backends (vLLM), vision-language models (VLMs), multimodal models, world modeling, state representations, mixture of experts (MoE), distributed training with Ray, DDP and FSDP, CUDA, large-batch optimization, GPU acceleration, multi-threading, FLOPs, parallel layout (data, tensor, pipeline parallelism), memory bandwidth, scaling laws, test-time vs. training-time compute, and distributed programming in HPC environments.

LLM Agents & Frameworks: Multi-agent systems, agentic AI workflows, planning, reasoning, agent design, agent training, agent fine-tuning, instruction tuning, agent memory (short-term and long-term), memory management, state tracking, tool-augmented reasoning, tool use, tool APIs, computer-use agents, action-taking agents, workflow automation, reliable task execution, multi-agent collaboration with feedback, multi-agent training, self-play, agent trajectories, agent-level parallelism, system optimization, KV-cache-aware inference, CLI tools, APIs, scripts, AutoGen, MCP (Model Context Protocol), LangChain, LlamaIndex, and LangGraph.

RAG, Search & Information Retrieval: Retrieval-augmented generation (RAG), hybrid search, semantic search, keyword search, dense retrieval, sparse retrieval, reranking, semantic indexing, vector embeddings, distributed information retrieval systems, search engines, knowledge graphs, indexing, ranking, query understanding, answering pipelines, enterprise search, retrieval reliability, relevance optimization, vector databases (FAISS), and evaluation frameworks such as RAGAS.

Machine Learning Libraries & Tools: PyTorch, TensorFlow, Keras, NumPy, pandas, scikit-learn, matplotlib, Hugging Face Transformers, vLLM, Weights & Biases, Vertex AI, GANs, representation learning, synthetic data generation, experiment tracking, benchmarking, and ML lifecycle tools.

Web Technologies: Django, Laravel, HTML, CSS, Bootstrap, Odoo, Django REST Framework, FastAPI, and Flask.

Databases, Cloud & DevOps: SQL, PostgreSQL, SQLite, Git, Docker, Neo4j, and cloud platforms including Azure, AWS, and GCP.

HONORS & AWARDS

- **3MT Best Paper Presentation Award**, Joint Conference on Digital Libraries (JCDL), Hong Kong — December 2024
- **USAII Global AI Hackathon 2026**, Got 3rd Place in Idea sharing and finalist main round. Virtual - June 2026
- **Ignite64 AI Hackathon 2026**, Got 2nd Place. Virtual - June 2026
- **IEEE Travel Grant**, IEEE International Conference on Data Science and Advanced Analytics (DSAA), San Diego, California — October 2024
- **SIGIR Student Travel Grant**, 2024 ACM/IEEE Joint Conference on Digital Libraries (JCDL), Hong Kong — December
- **Google GCP Grant Award**, Feb 2024 to Jan 2025, Dec 2025-Dec 2026
- **NIU Student Travel Grant**, December 2024 (for Attending JCDL 2024), September 2025 (for Attending IEEE eScience Chicago), November 2025 (for Attending EMNLP 2025)
- **NIU Outstanding Graduate Student Summer Fellowship**, Summer 2026
- **MMLS Travel Grant**, June 2026 (for Midwest Machine Learning Symposium at Purdue University)
- **NSF Travel Grant**, Sept 2026 (for Attending Join Conference at Digital Libraries (JCDL), Dallas, Texas)

OTHER

Reviewer

- ACL Workshop 2024, 2025, SDM 2025, IJDL 2025, 2026, ACM Computing Survey 2025, Springer Nature, WWW, AACL-IJCNLP

Program Committee Member

- Web of Science 2025, JCDL 2025, AACL 2025

Volunteer & Session Chair

- DSAA 2024, JCDL 2024, JCDL 2025

Speaker

- [NIU AI/ML Seminar]: September 2026: White box Attacks, Mechanistic Interpretability, and Defense Mechanisms on LLM. [Website](#)
- [NIU AI/ML Seminar]: February 2026: How to integrate Reinforcement Learning with LLM Agents. [Website](#)
- [NIU AI/ML Seminar]: September 2025: Adversarial Attacks and Defense Mechanisms on LLM. [Website](#)
- [NIU AI/ML Seminar]: May 2025: Limitations and Future Directions of Scientific Article. [Video](#)
- [NIU AI/ML Seminar]: April 2025: Evaluation system of Retrieval Augmented Generation. [Video](#)
- [NIU AI/ML Seminar]: February 2025: Large Language Models, How, why and how? [Video](#) [Website](#)
- Blog: Recent LLM Developments [ibrahimalazhar/LLM](#)

News Coverage

- NIU News Coverage for Attending MSLD 2026: [News](#)

Software Release

- A live website with Agentic AI Limitation Generation which performs better than frontier models (ChatGPT, Gemini, Llama, Mistral, and Qwen.) [Website](#)

Problem Solving

- Solved over 100 Problems from Leetcode, over 350 problems from Codeforces, UVa.

POSTERS

1. **Midwest Speech and Language Day 2025, Indiana:** Automated Extraction and Generation Limitations of Scientific Article, April, 2025. [Poster](#).
2. **Midwest Speech and Language Day 2026, Urbana:** LimAgents: A Multi Agent Framework for Research Limitations Generation.
3. **Greater Chicago Workshop 2025, Loyola University, Chicago:** Optimizing Database Performance in Retrieval-Augmented Generation with Adaptive Retrieval and Efficient Evaluation Methods, May, 2025. [Poster](#).
4. **Greater Chicago Workshop 2026, Northwestern University:** Scalable Multi-Agent RAG for Scientific Critique Generation at Corpus Scale.
5. **Midwest Machine Learning Symposium 2025, University of Chicago:** Optimizing Database Performance in Retrieval Augmented Generation with Adaptive Retrieval and Efficient Evalaution Methods.
6. **Midwest Machine Learning Symposium 2026, Purdue Univeristy:** SciLimAgents: An Offline Preference-Aligned Multi-Agent RAG Framework for Research Limitation Generation.

REFERENCES

Dr. Hamed Alhoori

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Md. Saiful Islam

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Shahjalal University of Science and Technology,

Bangladesh

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Dr. Sagnik Ray Choudhury

Assistant Professor - Computer Science
University of North Texas
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